

Tahsin Fuad Hasan

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SUMMARY

MS in Computer Science with 3+ years developing machine learning systems in Python, PyTorch, and C/C++ across computer vision, NLP, and wireless sensing. Designed an embedding-based retrieval model with privacy-preserving federated training (ACM TIOT 2026 submission), and a set-invariant attention mechanism for adaptive similarity learning under variable sensor conditions (AAAI 2026). Hands-on experience deploying models under real hardware constraints (GCP/Vertex AI training, embedded edge deployment).

EDUCATION

University at Buffalo, SUNY - MS in Computer Science and Engineering (Research Track) 2024 - Spring 2026

International Islamic University Malaysia - MS in Engineering, Mechatronics 2022 - 2023

International Islamic University Malaysia - BE in Mechatronics (Honours) 2017 - 2022

Relevant Courses: Deep Learning • Machine Vision • Robotics Algorithms • Data Intensive Computing • Digital Signal Processing • Embedded Systems

TECHNICAL SKILLS

ML/AI: PyTorch, TensorFlow, Scikit-learn, Embeddings, Attention Mechanisms, CNNs (ResNet, U-Net, MobileNet, DenseNet, YOLOv8n), LLMs (BERT/Hugging Face), NLP, Federated Learning, Predictive Modeling, Hyperparameter Tuning

Languages: Python (Pandas, NumPy, SciPy, Matplotlib), C/C++, Java, R, SQL, MATLAB

Tools: Git, Docker, GCP, Vertex AI, AWS, CI/CD (GitHub Actions), Linux, Jupyter, ROS, AI Dev Tools (Claude Code, Copilot)

PUBLICATIONS

- **Hasan, T. F.^{*}**, et al. *FedWiLoc: Privacy-Preserving WiFi Localization*. Submitted to IMWUT 2026. [Paper](#) | [Code](#)
- **Hasan, T. F.[†]**, et al. *Attending to Routers for Indoor Localization*. AAAI 2026. [Paper](#)
- Saadeldin, A., ..., **Hasan, T. F.** *Real-time Vehicle Counting Using YOLOv8n and DeepSORT*. TELKOMNIKA, 2024. [Paper](#)
- **Hasan, T. F.**, et al. *Water Quality Monitoring Using ML and IoT*. CNRE Journal, 2024. [Paper](#)

EXPERIENCE

Research Assistant | [WIRES Lab](#), University at Buffalo Jan 2024 - June 2026

Supervisor: Dr. Roshan Ayyalasomayajula

- Built ATLAS, a privacy-preserving indoor localization system that lets a user's device determine its own position from WiFi signals without exposing raw data to any server. The system uses federated learning, a split-computing architecture, and a custom loss function to generalize across unseen indoor environments without retraining.
- Developed data-cleansing and feature-extraction pipelines in Python and MATLAB to process over 100K data frames of raw Channel State Information (CSI) across 12 wireless deployments, debugging 2D AoA-ToF matrix transformations and geometric triangulation to extract, validate, and serialize continuous physical features into optimized HDF5 (.h5) datasets.
- Evaluated and benchmarked diverse deep learning architectures (U-Net, DenseNet, MobileNet and ResNet) in PyTorch before designing an optimized ResNet34 + MLP decoder topology that improved localization accuracy by 61.9% over 3 published baselines.
- Improved cross-environment generalization by 59% by designing a geometric loss function from first principles (trigonometric decomposition + differentiable triangulation) and reduced 99th-percentile outlier error by 39.4% through an attention mechanism for adaptive feature weighting. Both validated through systematic ablation studies.
- Reduced 99th-percentile outlier error by 39.4% by designing an attention mechanism that identifies and downweights degraded sensor inputs and improved performance at difficult locations by 45.5%.
- Trained models on GCP/Vertex AI and deployed to embedded hardware (328 MB, 369ms) under resource constraints. Created visualizations and evaluation reports to communicate findings to collaborators across 2 institutions.
- Maintained code quality through peer code reviews, automated testing, and CI/CD pipelines (GitHub Actions). Open-sourced the full codebase with documentation. Two first-author papers (ACM TIOT 2026 submission, AAAI 2026).

Teaching Assistant | University at Buffalo Jan 2024 - May 2026

- Taught SQL, relational databases, data structures (Java/C++ assignments), and algorithms across 3 semesters to 200+ students. Created instructional materials and provided detailed feedback on code and analytical work.

Research Assistant | International Islamic University Malaysia Sept 2022 - Dec 2023

- Built an IoT water-quality monitoring platform on Arduino UNO (ATmega328), integrating pH, turbidity, TDS, temperature, and dissolved-oxygen sensors, streaming real-time telemetry via MQTT to a Node-RED dashboard. Cleaned and processed 3k+ raw entries to 526 validated records; ran correlation analysis to identify pH and turbidity as primary predictors (50% correlation with potability) and benchmarked 6 ML algorithms, with Gradient Boosting reaching 99.3% accuracy. Published in CNRE Journal (2024).
- Co-developed a real-time detection and tracking system (YOLOv8n + DeepSORT) deployed on resource-constrained edge devices in Python and C++. Published in TELKOMNIKA (2024).

Engineering Intern | AT Systech Sdn. Bhd., Kuala Lumpur Jul - Oct 2021

- Supported deployment, testing, and commissioning of automated manufacturing equipment. Worked with PLC-based control systems (Allen Bradley) and engineering documentation (AutoCAD, SolidWorks).

PROJECTS

- **Misinformation Detection:** Fine-tuned BERT (Hugging Face) for multi-class text classification on LIAR2 dataset (89%+ accuracy). Cross-validated against LSTM.
- **PolarShift** (in progress): Developing an NLP pipeline using transformer embeddings to quantify changes in political discourse over time through topic modeling and semantic similarity analysis.
- **F1Tenth Autonomous Racing** (CSE 568): Autonomy stack in ROS with obstacle avoidance, particle filter localization, RRT planning, and sensor fusion.

ACTIVITIES

Presented research at SUNY AI Symposium (2026). Volunteered at ACM MobiSys (2025). VP of Bangladeshi GSA at UB (4+ events, 200+ attendees). President of MUN Club at IIUM (10+ awards).